Spring Research Project: Mentor Rubric

Mentor Name(s): Arjun Murlidhar

Mentee Name(s): [redacted]

Project Title: Finding Rogue Planets with Roman

In a short paragraph, explain the project and its learning objectives:

This project is aimed at introducing the mentees to the exciting field of microlensing and its potential to discover free-floating planets (FFPs) in the galaxy. The mentees will first go through some background material to learn the fundamentals of the technique of microlensing. They will then learn how to fit a basic (1 lens 1 source) microlensing model to data by working through a pre-written python notebook.

In the final part of the project, the mentees will use the code in the notebook to fit a large number of light curves from the 2026 Roman Microlensing Data Challenge (https://rges-pit.org/data-challenge/) and plot the distribution of the event timescales to identify FFP candidates in the data. The mentees can choose to work independently or as a team to complete this project. Most of the code required for this project is already written, and the mentees will only have to write code to create the final plot.

What are the deliverables? (i.e. plot/figure/image/equation mentee will reach)

The main deliverable is a plot of event timescales for the light curves in the data challenge and a list of FFP candidates. The mentees will also produce a list of light curve parameters for all the light curves fit as a byproduct.

Project Calendar: Indicate what the mentee will be expected to do each week. Work days are indicated in the left column. Include estimated time, resources, in/out of class work, and whether it is completed on their own or with you. The project should take 8-10 hours in total.

Dates:	Tasks:
Week 1 (2/10-2/16) Intro to Project	Introduction to microlensing, Roman, and rogue planets. HW: Go through introductory materials in the reading list
Week 2 (2/17-2/23) In Class Work Day	Discussion on microlensing and clarification of doubts from the reading, if any.
Week 3 (2/24-3/2) In Class Work Day	Learn about the single lens model. Download the python notebook and set it up.
Week 4 (3/3-3/8) Class Lecture	Complete set up of notebook. Fit a single lens model to data.
Spring Break (3/10-3/14)	None
Week 5 (3/17-3/23) In Class Work Day	Revise single lens models and fitting. Download light curves from the data challenge. HW: Write intro section for the poster
Week 6 (3/24-3/30) Class Lecture	Run the notebook on downloaded data. HW: write methods section for the poster
Week 7 (3/31-4/6) Class Lecture	Analyze results from the run, debug if necessary. HW: Create a histogram of event timescales for all the light curves
Week 8 (4/7-4/13) In Class Work Day	Identify FFP candidates from the distribution of event timescales. Discuss what the results imply for the upcoming Roman galactic bulge survey. HW: Work on poster
Week 9 (4/14-4/20) Poster Work Day	Work on poster
Week 10 (4/21) Present!	

Reading List: (Make sure material is level appropriate.)

To learn about microlensing:

Level 1:
Start Here:
https://science.nasa.gov/mission/roman-space-telescope/microlensing/
Watch the first 19 mins of:
https://www.youtube.com/watch?v=vuxb6jjg5vc
Read the first two sections of (Introduction to Microlensing and Point Lenses):
https://www.microlensing-source.org/concept/

Level 2:
If you are not thoroughly confused yet, and feel up for some more advanced discussion on the theory of microlensing (more math):
Read this: https://www.microlensing-source.org/tutorial/pspl/

Watch these lectures:
1) https://rges-pit.org/outreach_mini_ch1_video/
2) https://rges-pit.org/outreach_mini_ch2_video/

Read this review article:
https://www.annualreviews.org/content/journals/10.1146/annurev-astro-081811-125518
(Ask me for the PDF if you can't access it)

To learn about Roman:

Level 1:

https://science.nasa.gov/mission/roman-space-telescope/

https://science.nasa.gov/mission/roman-space-telescope/galactic-bulge-time-domain-survey/

To learn about rogue planets:

Level 1:

Read:
https://www.nasa.gov/missions/roman-space-telescope/new-study-reveals-nasas-roman-could-find-400-earth-mass-rogue-planets/
https://www.space.com/astronomy/exoplanets/astronomers-detect-rare-free-floating-exoplanet-10-000-light-years-from-earth

Watch the first 5 mins of:
https://www.youtube.com/watch?v=fz9RUGlqUZo

Level 2:
Relevant scientific papers:
https://arxiv.org/abs/2006.10760
https://arxiv.org/abs/2601.00057

Resource List: (This should include lab space, instruments, and software. Note that mentees are not allowed to complete research projects that require safety training operation of lab equipment. If you feel this is detrimental to your project, please petition Polaris leadership.)

Notebooks:
https://github.com/AmberLee2427/TheMicrolensersGuideToTheGalaxy/tree/main
https://github.com/rges-pit/data-challenge-notebooks/blob/main/AAS%20Workshop/Session%20B%3A%20Single%20Lens%20%26%20Pipelines/Single_Lens_Pipeline.ipynb

Other Notes: